

**JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
SPRDL1-23-R-0113**

1. Contracting Agency: Defense Logistics Agency Land-Warren. (DLA -WRN)

2. Description of Action: DLA-WRN proposes to award a three-year Firm-Fixed-Price, Indefinite Delivery Indefinite Quantity contract to Kidde Technologies, Inc., Commercial and Government Entity (CAGE) 05BU0. The contract will procure National Stock Numbers (NSNs) 4210-01-566-2417, Fire Extinguisher Assembly, NSN 4210-016-61-2573, Extinguisher, Fire, and NSN: 4210-01-645-8356, Fire Extinguisher using Army Working Capital Funds.

3. Description of Supplies/Services: The supplies under this proposed contract are as follows:

PRON	EH2L0080EH	EH2L0203EH	EH2L0187EH
Item Name	Extinguisher Assembly, Fire	Extinguisher, Fire	Extinguisher, Fire
NSN	4210-01-566-2417	4210-01-645-8356	4210-01-661-2573
Manufacturer Part Number	421625	423245	423244
Government Part Number	12496698	12542354-2	12542354-4
CAGE Code	05BU0	05BU0	05BU0
Contract Type	Long Term Contract	Long Term Contract	Long Term Contract
Unit Cost			
Yearly Usage			
Minimum Guaranteed	135 each	11 each	10 each
Maximum Quantity	1,444 each	79 each	50 each
Minimum Contract Cost			
Maximum Contract Cost			
TOTAL MIN			
TOTAL MAX			

NSN 4210-01-566-2417, Government part number (P/N) 12496698, Fire Extinguisher Assembly, is used on the Bradley Family of Vehicles (FoV) to suppress fires within the crew compartment. The deliverables for this contract are a minimum quantity of 135 each, with a three-year maximum quantity of 1,444 each.

NSN 4210-01-645-8356, Government P/N 12542354-2, Fire Extinguisher, is utilized on the Self-Propelled Howitzer System (SPHS) FoV to

suppress fires within the crew compartment. The deliverables for this contract are a minimum quantity of 11 each, with a three-year maximum quantity of 79 each.

NSN: 4210-01-661-2573, Government P/N 12542354-4, Fire Extinguisher, is utilized on the SPHS FoV to suppress fires within the engine compartment. The deliverables for this contract are a minimum quantity of 10 each, with a three-year maximum quantity of 50 each.

Purchase Request Order Number (PRON) EH2L0080EH, EH2L0203EH and EH2L0187EH are assigned to this procurement. The estimated total maximum value of this contract is \$ [REDACTED].

Acquisition Method Codes (AMC): 3, 4

3 - Acquire, for the second or subsequent time, directly from the actual manufacturer.

- NSN 4210-01-566-2417
- NSN 4210-01-661-2573

4- Acquire, for the first time, directly from the actual manufacturer.

- NSN 4210-01-645-8356

Acquisition Method Suffix Code (AMSC): B

- NSN 4210-01-566-2417
- NSN 4210-01-661-2573
- NSN 4210-01-645-8356

B - This part must be acquired from a manufacturing source(s) specified on a source control or selected item drawing as defined by the current version of DoD-STD-100. Suitable technical data, Government data rights, or manufacturing knowledge are not available to permit acquisition from other sources, nor qualification testing of another part, nor use of a second source part in the intended application. Although, by DoD-STD-100 definition, altered and selected items shall have an adequate Technical Data Package (TDP), data review discloses that required data or data rights are not in Government possession and cannot be economically obtained. If one source is available, AMCs 3, 4, or 5 are valid. If at least two sources exist, AMCs 1 or 2 are valid.

4. Authority Cited: The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204 (a)(1) as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Reason for Authority Cited: The Automatic Fire Extinguisher System (AFES) is an interconnected network of Fire Extinguisher Assemblies that disburse Halon and/or Heptafluoropropane fire suppression agent throughout the vehicle's crew and engine compartments via specific nozzles locations. The AFES is controlled by a single

electronics panel which is connected to fire sensors throughout the vehicle. The AFES utilizes the three NSNs listed above and is used in the Bradley and SPHS FoV. The AFES is a critical safety system that automatically senses and extinguishes fires that would endanger the life of the crew. The AFES is required to automatically detect a fire and dispense extinguishing agent within 250 milliseconds. Specific to the SPHS FoV, the AFES shall extinguish a fire in 12 seconds within the engine compartment and in 10 seconds within the crew compartment. Kidde Technologies, Inc. has demonstrated that Vendor P/N 423244 and 423245 have met all system level performance requirements, as defined by Section 3.0 of the AFES Component Specification, drawing number 12578486. Kidde Technologies, Inc. also demonstrated that Vendor P/N 421625 has met the Fire Extinguisher System, Sections 3.1.5.8 of 12465518 for the Bradley FoV, including integration within the vehicle. Kidde Technologies, Inc. has developed an appropriately sized Fire Extinguisher Assembly that works within the AFES system. The Fire Extinguisher Assemblies are designed to hold a specific amount of extinguishing agent at a specific pressure in order to disperse an adequate amount of agent during a fire. These Fire Extinguisher Assemblies have a specific form/fit that allows them to mate up to the existing AFES. Kidde Technologies, Inc. developed, tested, and validated each Fire Extinguisher Assembly within the AFES.

The Government owns two source-controlled drawings that define some of the Fire Extinguishers envelope dimensions and performance requirements. The drawings are not fully defined and are insufficient for manufacturing and/or qualifying an alternate vendor's item. The drawings do not identify any materials and do not define any down components. Kidde Technologies, Inc. is the only approved source of supply identified on each of these drawings. Kidde Technologies, Inc. is the sole owner of the fully defined technical data that is needed to produce vendor part numbers 421625, 423245, and 423244. As a result, the Government does not have proprietary rights, nor the pertinent technical data to produce a competitive TDP. The Government does not own the necessary intellectual property data rights to enable competitive re-procurement.

Kidde Technologies, Inc. has demonstrated the unique and proven manufacturing capabilities to establish and maintain the processing control required to repetitively produce this item. The use of incorrect material and/or fabrication of this item could result in the vehicle being unsafe or prohibited from being fielded if the AFES is inoperable.

Reverse engineering is the only recourse the Government could entertain to increase competition through development of a competitive TDP for the various fire extinguishers utilized within the AFES. The complex nature of the AFES hardware requires significant design validation testing and vehicle system level testing.

Ground Vehicle System Center (GVSC) Engineering collaborated with an auxiliary systems Subject Matter Expert (SME) from the SPHS Program Office to define the reverse engineering effort and associated cost. The SME estimate is based on previous experience evaluating the initial development costs of these items and other comparable AFES. The required stages of reverse engineering and associated cost by NSN are described below and summarized in the associated table.

Developing the hardware design and requirements is estimated to be \$100,000 in cost for each of the three items being procured. Integrating and fabricating P/N 421625, Fire Extinguisher Assembly hardware onto the Bradley FoV for the required testing is estimated to be \$1M. Integrating and fabricating Fire Extinguisher P/Ns 423244 and 423245 onto the SPHS FoV for the required testing is estimated to be \$500,000 each.

Reliability and Maintainability (RAM) testing, which includes both component and vehicle level testing, for P/N 421625, Fire Extinguisher Assembly hardware on the Bradley FoV was calculated by the United States Government (USG) testing group as follows: RAM testing consisting of 5,000 miles of mobility test at a cost of \$2M and full Electromagnetic Interference (EMI) testing at a cost of \$325,000, for a total of \$2.325M. RAM testing for P/Ns 423244 and 423245 on the SPHS FoV is calculated to be \$400,000, consisting of 1,000 of miles mobility testing. The reduced mileage required for mobility testing is due to lower complexity of the parts being procured. These items also do not require EMI testing, as there are no electronic components.

The USG does not own the TDP for the of P/N 421625, nor the TDPs for P/Ns 423245 and 423244. Developing and validating a TDP for P/N 421625 is estimated to be \$250,000 in cost and includes the multiple down-components within the assembly. Developing and validating TDPs for P/Ns 423244 and 423245 is estimated to be \$100,000 for each item due to the absence of additional system down-component parts needing test.

Reverse Engineering Activity	Extinguisher Assembly, Fire	Extinguisher Fire	Extinguisher Fire
Manufacturer Part Number	421625	423245	423244
Requirements Development & Hardware Design	\$100,000	\$100,000	\$100,000
Integration & Fabrication/Installation	\$1,000,000	\$500,000	\$500,000
Testing (including applicable component level, RAM, and EMI)	\$2,325,000	\$400,000	\$400,000
Develop & Validate TDP	\$250,000	\$100,000	\$100,000
Total Cost for Reverse Engineering	\$3,675,000	\$1,100,000	\$1,100,000
Unit Cost			
Annual Demand Rate			
Assumed Savings			
Annual Savings			
Return on Investment (ROI)			

Total Reverse Engineering Costs are calculated to be \$3,675,000 for P/N: 421625 and \$1,100,000 each for P/Ns 423245 and 423244. The Government would not be able to recover within a reasonable amount of time, through subsequent competition, the duplication of cost involved in reverse engineering any of the items to create new and competitive TDPs. At a current average usage rate of [REDACTED] Fire Extinguisher Assemblies, Vendor P/N 421625 each year, at an anticipated unit cost of \$ [REDACTED], and assuming

■ % savings achieved through competition, the \$ ■ in annual savings would take over ■ to offset the estimated \$3,675,000 cost of reverse engineering the item and validating the TDP.

At a current average usage rate of ■ Fire Extinguishers', Vendor P/N 423245, per year at an anticipated unit cost of \$ ■ and assuming ■ % savings achieved through competition, the \$ ■ in annual savings would take over ■ to offset the estimated \$1,100,000 cost of reverse engineering the item and validating the TDP.

At a current average usage rate of ■ Fire Extinguishers', Vendor P/N 423244, per year, at an anticipated unit cost of \$ ■ and assuming ■ % savings achieved through competition, the \$ ■ in annual savings would take over ■ to offset the estimated \$1,100,000 cost of reverse engineering the item and validating the TDP.

6. Efforts to Obtain Competition:

- a. Effective competition: Due to the reasons cited in section 5 above, the Government is not pursuing competition for this item.
- b. Subcontracting competition: The solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages-- Subcontracting Plan"; and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

SOLE SOURCE: Notices required by FAR 5.201 will be posted.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 13 October 2022, was open for 48 days and closed 30 November 2022 with one response received. New Material Supply Co, (CAGE 1J8D2) responded 25 November 2022 with a capability statement. Engineering reviewed the statement and determined that the vendor's sourced manufacturer of the fire extinguishers did not have the capability to satisfy the Government's requirement of automatic fire detection and dispelling capabilities. The vendor was referred to the Army's Source Approval Request (SAR) Program via email 28 November 2022.

A web-based market research was completed by Ground Vehicle Systems Center (GVSC) Engineering in August 2022. No Off-the-Shelf readily available Automatic Fire Extinguisher System were identified using various search engines, such as Google and BING, for matching technical capabilities in detecting and dispelling fires occurring inside vehicle crew compartment and the engine compartment. Amerex Fire Corporation and Blackhawk Industrial are potential vendors with history of producing fire extinguisher applications; however, they or any new vendor would need to complete the

validation process consisting of extensive analysis, prototyping, and in-vehicle testing on the Bradley and SPHS FoV, as referenced in section 5 properly meet performance specifications with a new supplier's component. While other sources likely capable of developing this component exist; however currently, there are no other companies besides Kidde Technologies, Inc that have completed testing and are proven to possess this capability. Extensive analysis, prototyping, and vehicle testing would need to be done to ensure that the Bradley and SPHS FoV will properly meet performance specifications with a new supplier's component. As described above in section 4, the cost of such analysis, prototyping, and testing would far exceed any cost savings realized by adding another source of supply.

The Defense Contract Management Agency (DCMA) Commercial item Database (CID) was consulted on 31 October 2022 for NSNs 4210-01-661-2573, 4210-01-566-2417 and 4210-01-645-8356. Formal DCMA Commercial Determinations were available for NSNs 4210-01-661-2573 and 4210-01-645-8356. The third NSN, 4210-01-566-2417, did not have a formal CID from DCMA, however this product was described as commercial by vendor Kidde in email dated 7 November 2022, with the same rationale as reviewed and endorsed by DCMA for NSN 4210-01-661-2573 and 4210-01-645-8356. NSN 4210-01-566-2417 is of a product type that has been deemed commercial in the past and has also been procured commercially under the following prior contracts: SPRDL1-20-C-0142, SPRDL1-20-D-0063, SPRDL1-15-C-0094, SPRDL1-13-C-0126 and SPRDL1-12-C-0008.

Summary: As a result of the market research conducted, it was concluded that these products are used, sold/leased, or offered for sale to the public and are available in the commercial marketplace with or without modifications pursuant to the definition of a commercial product at FAR 2.101. These products do not meet the definition of a COTS product at FAR 2.101.

9. Interested Sources: To date, no additional sources have expressed interest in this item other than those as described above.

10. Other Facts: Procurement History:

SPRDL1-20-D-0063, DLA Land Warren
Awarded 16 March 2020 KIDDE TECHNOLOGIES (CAGE 05BU0)
NSN 4210-01-566-2417
Total Value \$1,656,802.44
FAR 6.302- I, "Only one responsible source and no other supplies or services will satisfy agency requirements."

SPRDL1-20-C-0142, DLA Land Warren
Awarded 26 May 2020 KIDDE TECHNOLOGIES (CAGE 05BU0)
NSN 4210-01-566-2417
Total value \$2,325,146.10
FAR 6.302- I, "Only one responsible source and no other supplies or services will satisfy agency requirements."

SPRDL1-15-C-0094, DLA Land Warren

Awarded 26 February 2015 KIDDE TECHNOLOGIES (CAGE 05BU0)

Total Value \$1,159,124

NSN 4210-01-566-2417

FAR 6.302- I, "Only one responsible source and no other supplies or services will satisfy agency requirements."

SPRDL1-13-C-0126, DLA Land Warren

Awarded 26 July 2013 KIDDE TECHNOLOGIES (CAGE 05BU0)

Total Value \$317,440

NSN 4210-01-566-2417

FAR 6.302- I, "Only one responsible source and no other supplies or services will satisfy agency requirements."

SPRDL1-12-C-0008, DLA Land Warren

Awarded 06 October 2011 KIDDE TECHNOLOGIES (CAGE 05BU0)

Total Value \$562,440

NSN 4210-01-566-2417

FAR 6.302- I, "Only one responsible source and no other supplies or services will satisfy agency requirements."

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]
Title: Division Chief, Sustainment Engineering
Email: [REDACTED]
Signature:

8/17/2023

X [REDACTED] i 8/17/23

Signed by: [REDACTED]

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]
Title: Division Chief, Sourcing
Email: [REDACTED]
Signature:

8/18/2023

X [REDACTED]

Signed by: [REDACTED]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and/ or price analysis. The contractor will not be required to submit certified cost and pricing data as this item has been determined to be a commercial product.

Name: Eric Tucker
Title: Contracting Officer, DLA Land Warren, ZGAB
Email: eric.tucker@dla.mil
Signature:

8/21/2023
X Eric Tucker
Signed by: TUCKER.ERIC.E [REDACTED]

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Eric Tucker
Title: Contracting Officer, DLA Land Warren, ZGAB
Email: eric.tucker@dla.mil
Signature:

8/21/2023
X Eric Tucker
Signed by: TUCKER.ERIC.E [REDACTED]

**Approval Authority for Acquisitions Equal to or Greater Than \$750,000
But less than \$15M**

**APPROVAL OF JUSTIFICATION FOR
OTHER THAN FULL AND OPEN COMPETITION**

Based on the forgoing Justification, I hereby approve procurement of the following:

- a. NOUN/NSN:
FIRE EXTINGUISHER, NSN 4210-01-661-2573
FIRE EXTINGUISHER ASSEMBLY, 4210-01-566-2417
FIRE EXTINGUISHER, 4210-016-45-8356

- b. Minimum Quantity:
NSN: 4210-01-661-2573 - 10 each
NSN 4210-01-566-2417- 135 each
NSN 4210-016-45-8356 - 11 each

Maximum Quantity:
NSN: 4210-01-661-2573 - 50 each
NSN 4210-01-566-2417- 1,444 each
NSN 4210-016-45-8356 - 79 each

- c. Basic Estimated Value: \$ [REDACTED]
d. Option Percentage: N/A- IDIQ
e. Quantity with Option: N/A- IDIQ
f. Total Estimated Maximum Value: \$ [REDACTED]

This action will be awarded under other than full and open competition pursuant under the authority of 10 United States Code (USC) 3204(a)(1) and subject to the availability of funds. This approval is subject to the provision that the property herein described has otherwise been authorized for acquisition.

SHEARER.MAR
X Y.E [REDACTED]
Digitally signed by
SHEARER.MARY
Date: 2023.08.30 07:49:59 -04'00'

Competition Advocate